PHYS 143 – Mechanics and Thermal Physics

# Impulse, Momentum and Collisions

*How do forces acting over a time change motion,*
*and what stays the same when objects interact?*

Hope Donglo

Office 24, Department of Physics, University of Ghana

# Hope Donglo

Room 24, Department of Physics

hdonglo@ug.edu.gh


Office hours: Wednesdays, 10:00 to 12:00

Personal page: hhdonglo.github.io/phys143.html

## By the end of this week, you should be able to:

1. Calculate momentum and impulse, and relate them with the impulse–momentum theorem.
2. Apply conservation of momentum to isolated systems.
3. Solve one- and two-dimensional collision problems, and classify collisions as elastic or inelastic.
4. Use the coefficient of restitution, combine collisions with energy methods, and find a centre of mass.

### Central question

How do forces acting over a time change motion, and what stays the same when objects interact?

### The thread of the week

**force over time** $\rightarrow$ **impulse** $\rightarrow \Delta\vec{p} \rightarrow$ **isolated system:** $\vec{P}$ **stays constant** $\rightarrow$ **collisions.** Week 4 summed force over distance; this week we sum force over time.

**Two contacts: A** (Tuesday) **DEFINE** momentum and impulse, then **CONSERVE** momentum and sort collisions by their kinetic energy; **B** (Wednesday) **SOLVE** collisions along a line and in two dimensions, then **EXTEND** to restitution, collision then energy, and the centre of mass.

**Reused:** Lecture 3 ($\vec{p} = m\vec{v}$, $\sum \vec{F} = \mathrm{d}\vec{p}/\mathrm{d}t$, third law), Lecture 4 (work–energy, $K$), Lecture 5 ($U$, energy conservation, friction), Lecture 1 components. **New:** impulse, conservation of $\vec{P}$, collision types, $e$, $\vec{P} = M\vec{v}_{\mathrm{cm}}$. **Later:** Lectures 8, 10, 11, 12.

**Reading:** Young and Freedman, Chapter 8, §8.1–8.5. **Labels:** **CORE** you should be able to do this; **PRACTICE** develops the skill; **REFERENCE**

# Where we start: a five-minute check

**Not marked. Try each, then check.**

1. A $0.50\,\mathrm{kg}$ ball moves at $4.0\,\mathrm{m/s}$ east and then at $2.0\,\mathrm{m/s}$ west. Change in its velocity?
2. A constant $12\,\mathrm{N}$ force acts for $3.0\,\mathrm{s}$ on a $4.0\,\mathrm{kg}$ object at rest. Final speed?
3. State Newton's third law. On which bodies do the two forces act?
4. Area under a force graph that rises linearly from $0$ to $20\,\mathrm{N}$ over $0.10\,\mathrm{s}$?
5. Kinetic energy of a $2.0\,\mathrm{kg}$ object moving at $3.0\,\mathrm{m/s}$?
6. Resolve $10\,\mathrm{m/s}$ at $37^\circ$ above the $+x$ axis into components.

**Answers**

**1.** East $+$:
$\Delta v = -2.0 - 4.0 = -6.0\,\mathrm{m/s}$

**2.** $a = 12/4.0 = 3.0\,\mathrm{m/s^2}$;
$v = at = 9.0\,\mathrm{m/s}$

**3.** Equal and opposite, on *different* bodies.

**4.** $\frac{1}{2}(0.10)(20) = 1.0\,\mathrm{N\,s}$

**5.** $K = \frac{1}{2}(2.0)(3.0)^2 = 9.0\,\mathrm{J}$

**6.** $v_x = 10\cos 37^\circ = 8.0$,
$v_y = 10\sin 37^\circ = 6.0\,\mathrm{m/s}$

Items 1 and 3 are Weeks 2 and 3, item 4 is a graph area, item 5 is Lecture 4, item 6 is Week 1. All come back this week.

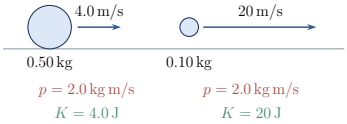
## Contact A: Define, then Conserve

*What are momentum and impulse,*
*and when is momentum conserved?*

# Which ball is easier to catch?

## Lecture 4

A force acting over a *distance*: work, $\Delta K$.




## Key idea

Same mass $\times$ speed. Very different $\frac{1}{2}mv^2$.

## Predict first

Predict: which ball is easier to catch? Does the hand need more *time* or more *distance* for one of them?

## Lecture 7

A force acting over a *time*: impulse, $\Delta\vec{p}$. *What does that change?*

## Opening question

A $0.50\,\mathrm{kg}$ ball moves at $4.0\,\mathrm{m/s}$ and a $0.10\,\mathrm{kg}$ ball at $20\,\mathrm{m/s}$. Your hand exerts the same constant force to stop either. Which ball is easier to catch? Compare the time and the distance needed to stop each.

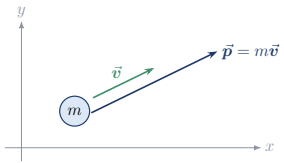
**But what if...** we try numbers? Which of $mv$ and $\frac{1}{2}mv^2$ sets the time, and which the distance?

# The vocabulary of momentum and impulse

CORE

| Quantity | What it is | What it tells you | SI unit | How to picture it |
|---|---|---|---|---|
| **Momentum $\vec{p}$** | the product of a particle's mass and velocity, $\vec{p} = m\vec{v}$; a vector along $\vec{v}$ | how much motion, and which way; its rate of change is the net external force | $\mathrm{kg\,m/s}$ | an arrow along $\vec{v}$ |
| **Net external force $\sum \vec{F}$** | the vector sum of the forces that objects outside the system exert on it | what can change the momentum | N | a free-body diagram |
| **Impulse $\vec{J}$** | the net external force multiplied by the time it acts; in general, its time integral | the change in momentum, $\vec{J} = \Delta\vec{p}$ | $\mathrm{N\,s} = \mathrm{kg\,m/s}$ | area under the $F$–$t$ graph |
| **Average force $\vec{F}_{\text{av}}$** | the constant force that gives the same impulse in the same time | the size of a collision force that varies | N | a flat line at the same area |

**Reading the table.** Momentum asks: *how much motion, and which way?* Impulse asks: *how much did the net external force change it?*




## Key idea

The momentum $\vec{p}$ points the same way as the velocity $\vec{v}$.

## Predict first

The same car moves at $20\,\mathrm{m/s}$ east, then at $20\,\mathrm{m/s}$ west. Which of $\vec{p}$ and $K$ changes? *Answer:* $\vec{p}$ reverses; $K$ does not. $p$ is a vector, $K$ a scalar.

## Rule: momentum and the second law

$$\vec{p} = m\vec{v} \tag{1}$$

$$\sum \vec{F} = \frac{\mathrm{d}\vec{p}}{\mathrm{d}t} \tag{2}$$

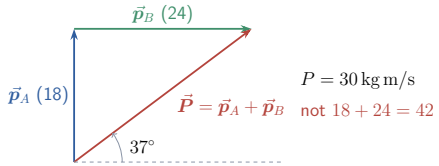
**Define.** $m$ is the mass (kg), $\vec{v}$ the velocity (m/s). $\vec{p}$ is the momentum, a **vector**, unit $\mathrm{kg\,m/s}$, with components $p_x = mv_x$, $p_y = mv_y$. $\sum \vec{F}$ is the net external force (N).

**Explain.** (2) says the net force on a particle is the rate of change of its momentum; for constant mass, $m\,\mathrm{d}\vec{v}/\mathrm{d}t = m\vec{a}$. A truck and a bicycle at the same speed: the truck has far more momentum, so the same force needs far more time to stop it.

*Reuse:* Lecture 3 (4a), (4b)

*eg. a crash test: the airbag must remove the occupant's momentum.*

# Momentum adds like a vector




## Key idea

Two perpendicular momenta add as a right-angled triangle, not by adding magnitudes.

## Predict first

$p_A = 18$ upward and $p_B = 24$ to the right ($\mathrm{kg\,m/s}$). Is the total $42\,\mathrm{kg\,m/s}$? Smaller or larger? *Answer:* smaller, $30\,\mathrm{kg\,m/s}$ (below).

## Rule: total momentum of a system

$$\vec{P} = \vec{p}_A + \vec{p}_B + \vec{p}_C + \cdots \qquad (3)$$

**Define.** $\vec{P}$ is the total momentum of a system of particles A, B, C, ... ($\mathrm{kg\,m/s}$): the **vector sum** of their momenta.

**Explain.** Add components: $P_x = p_{Ax} + p_{Bx} = 24$, $P_y = p_{Ay} + p_{By} = 18$. Then

$$P = \sqrt{24^2 + 18^2} = 30\,\mathrm{kg\,m/s}$$

$$\theta = \tan^{-1}\tfrac{18}{24} = 37^\circ \text{ above the } +x \text{ axis (quadrant I)}$$

Adding the magnitudes, $18 + 24 = 42$, is wrong: it works only when the momenta point the same way. *Reuse:* Lecture 1, components and reconstruction

## Rule: impulse–momentum theorem

$$\vec{\boldsymbol{J}} = \int_{t_1}^{t_2} \sum \vec{\boldsymbol{F}}\, \mathrm{d}t = \vec{\boldsymbol{F}}_{\mathrm{av}}\, \Delta t \qquad (4)$$

$$\vec{\boldsymbol{J}} = \Delta\vec{\boldsymbol{p}} = \vec{\boldsymbol{p}}_2 - \vec{\boldsymbol{p}}_1 \qquad (5)$$

**Define.** $\vec{\boldsymbol{J}}$ is the impulse of the net external force from $t_1$ to $t_2$ (N s), a **vector**, along $\sum \vec{\boldsymbol{F}}$ when that is constant. $\Delta t = t_2 - t_1$ (s). $\vec{\boldsymbol{F}}_{\mathrm{av}}$ is the average net external force (N). $\vec{\boldsymbol{p}}_1$, $\vec{\boldsymbol{p}}_2$ are the momenta at $t_1$ and $t_2$.

**Explain.** Start from (2) and integrate: $\int \sum \vec{\boldsymbol{F}}\, \mathrm{d}t = \int \mathrm{d}\vec{\boldsymbol{p}} = \vec{\boldsymbol{p}}_2 - \vec{\boldsymbol{p}}_1$. A constant force gives $\vec{\boldsymbol{J}} = \sum \vec{\boldsymbol{F}}\, \Delta t$. Unit: $\mathrm{N\,s} = \mathrm{kg\,m/s}$. Impulse is *not* a force.

*Reuse:* Lecture 3 (4b)

## Predict first

A constant $12\,\mathrm{N}$ force acts for $3.0\,\mathrm{s}$ on a $4.0\,\mathrm{kg}$ object at rest (check, question 2). Predict $J$ and the final speed.

$J = F\Delta t = (12)(3.0) = 36\,\mathrm{N\,s}$; $\Delta p = 36$, so $v = 36/4.0 = 9.0\,\mathrm{m/s}$, as from $a = 3.0\,\mathrm{m/s^2}$.

## Common error

Impulse is not a force: it is a force *multiplied by a time*, and equals the change in momentum.

## Valid for

any net external force: (5) is exact. Only the short form $\sum \vec{\boldsymbol{F}} \Delta t$ needs a constant force.

# Two theorems, two questions




$K$ gained $= \sum \vec{F} \cdot \vec{s}$ (over distance)

$\vec{p}$ gained $= \sum \vec{F}\,\Delta t$ (over time)

## Key idea

One force, two sums: over the distance it gives $\Delta K$, over the time it gives $\Delta \vec{p}$.

## Which question?

**Work–energy:** force over a *distance*, work $W$ (scalar), changes $K$: a speed or a distance.
**Impulse–momentum:** force over a *time*, impulse $\vec{J}$ (vector), changes $\vec{p}$: a velocity, a time or a collision force.

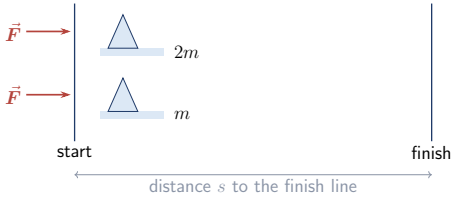
## Opening question, answered

**Momentum:** $p = mv = 2.0\,\mathrm{kg\,m/s}$ for *both* balls. The same force then needs the **same time**, $t = p/F$, from (5).

**Kinetic energy:** $\frac{1}{2}mv^2$ is $4.0\,\mathrm{J}$ for the $0.50\,\mathrm{kg}$ ball and $20\,\mathrm{J}$ for the $0.10\,\mathrm{kg}$ ball. The force removes it by work, $Fd = K$, so $d = K/F$ is **five times longer** for the small, fast ball.

## Answer

The $0.50\,\mathrm{kg}$ ball is easier to catch: your hand and arm are pushed back one fifth as far. Both theorems rest on Newton's laws; they are *integral* principles, while $\sum \vec{F} = \mathrm{d}\vec{p}/\mathrm{d}t$ holds at each instant.




**Key idea**

Same force, same distance: same $K$.
Different time, so different $p$.

**Example 1**

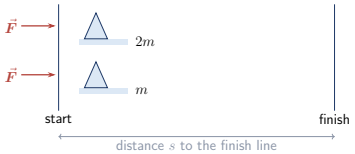
Two wind-powered carts of masses $m$ and $2m$ race on a frictionless horizontal track. Each has an identical sail, so the wind exerts the same constant force $F$ on each. Both start from rest and cross the finish line a distance $s$ away. Which cart crosses the line with the greater momentum, and why? How do their kinetic energies compare?

**Predict first**

Which cart has more kinetic energy at the finish? Which has more momentum? Which takes longer?

# Example 1: two sail carts, momentum and kinetic energy, solution

CORE




**Key idea**

Same force, same distance: same $K$. Different time, so different $p$.

**IDENTIFY**

Given: masses $m$, $2m$; the same $F$ and $s$; both from rest. Find: which has the greater $p$; compare $K$.

**SET UP**

The work is the same, $W = Fs$, so $K = Fs$ for both. *Reuse:* Lecture 4 (5) Momentum equals the impulse from rest, (5): $p = F\Delta t$.

**EXECUTE**

Equation: $K = \frac{1}{2}mv^2 = p^2/2m$, so $p = \sqrt{2mK} = \sqrt{2mFs}$

Substitute: $p_m = \sqrt{2mFs}$; $\quad p_{2m} = \sqrt{2(2m)Fs} = \sqrt{2}\sqrt{2mFs}$

Ratio: $\boxed{p_{2m}/p_m = \sqrt{2} = 1.41}$

**EVALUATE**

Same $K$, but $1.41$ times the momentum: the $2m$ cart accelerates more slowly ($a = F/2m$), takes $\sqrt{2}$ times as long (hint: solve for $t$, $s = \frac{1}{2}at^2$)

# Concept check: what is impulse?

## Decide first, with a reason for each, then check

1. A ball hits a wall at $10\,\mathrm{m/s}$ and bounces straight back at $10\,\mathrm{m/s}$. Is the change in its momentum zero?
2. Two balls have the same momentum. Does the heavier ball have the greater kinetic energy?
3. Is impulse a force? In which direction does it point?
4. The same impulse is given in $0.010\,\mathrm{s}$ and in $0.10\,\mathrm{s}$. Which needs the larger average force?

## Answers

**1. No.** $\Delta p = m(-10) - m(+10) = -20m$: reversing the velocity doubles the change, it does not cancel it.
**2. No.** $K = p^2/2m$: for the same $p$ the *lighter* ball has the larger $K$.
**3. No.** It is a force times a time, a vector along the net external force and equal to $\Delta\vec{p}$.
**4. The $0.010\,\mathrm{s}$ one.** $F_{\mathrm{av}} = J/\Delta t$: ten times the force.

## METHOD – Using $\vec{J} = \Delta\vec{p}$

**1. AXES:** choose a positive direction (or $+x$ and $+y$) before assigning any signs. *Reuse:* Lecture 2, signs
**2. STATES:** write $\vec{p}_1 = m\vec{v}_1$ and $\vec{p}_2 = m\vec{v}_2$ with signs, in components if the motion is in two dimensions.
**3. IMPULSE:** $\vec{J} = \vec{p}_2 - \vec{p}_1$, final minus initial, one equation per axis.
**4. AVERAGE FORCE:** $F_{\text{av}} = J/\Delta t$; direction along $\vec{J}$.
**5. NET FORCE:** $\vec{J}$ is the impulse of the *net* external force: include weight only if it matters over $\Delta t$.
**WATCH OUT:** reversing the velocity changes $p$ by $2mv$, not zero; the average force is not the peak force.
**CHECK:** does $\vec{J}$ point the way the force must have pushed?

## Which theorem?

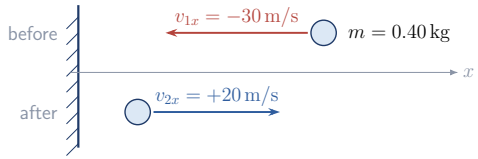
**Impulse–momentum (5):** a *velocity, a time, or a force during a short interaction.*

**Work–energy:** a *speed or a distance* when the force varies with position.

## Decide: which theorem?

(a) Contact time of a bat on a ball.
(b) Stopping distance of a skidding car.
(c) Speed of a ball after a kick, given the force and its duration.

**Answers:** (a) impulse, (b) work–energy, (c) impulse.




## Key idea

Choose $+x$ away from the wall. The ball's velocity changes sign.

## Example 2

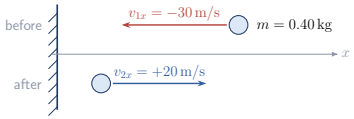
A boy throws a $0.40\,\mathrm{kg}$ ball against the brick wall of a school compound. It hits the wall moving horizontally to the left at $30\,\mathrm{m/s}$ and rebounds horizontally to the right at $20\,\mathrm{m/s}$. (a) Find the impulse of the net force on the ball. (b) The ball is in contact with the wall for $0.010\,\mathrm{s}$: find the average horizontal force the wall exerts on it.

## Predict first

Which way does $\vec{J}$ point? Is $F_{\mathrm{av}}$ larger or smaller than the ball's weight, $3.9\,\mathrm{N}$?

# Example 2: a ball hits a wall, solution

CORE




**Key idea**

Choose $+x$ away from the wall.
The ball's velocity changes sign.

**IDENTIFY**

Given: $m = 0.40\,\text{kg}$, $v_{1x} = -30\,\text{m/s}$, $v_{2x} = +20\,\text{m/s}$, $\Delta t = 0.010\,\text{s}$.
Find: $J_x$ and $(F_{\text{av}})_x$.

**SET UP**

$+x$ to the right, away from the wall. The wall's force is far larger than the weight over $0.010\,\text{s}$, so it is the net force. Use (5) and (4).

**EXECUTE**

(a) Equation: $J_x = p_{2x} - p_{1x} = m(v_{2x} - v_{1x})$
Substitute: $J_x = (0.40\,\text{kg})\,[(+20\,\text{m/s}) - (-30\,\text{m/s})]$
$p_{1x} = -12\,\text{kg m/s}$, $p_{2x} = +8.0\,\text{kg m/s}$ $\boxed{J_x = +20\,\text{N s}}$
(b) Equation: $J_x = (F_{\text{av}})_x\,\Delta t$, so $(F_{\text{av}})_x = J_x/\Delta t$
Substitute: $(F_{\text{av}})_x = \dfrac{20\,\text{N s}}{0.010\,\text{s}}$ $\boxed{(F_{\text{av}})_x = 2000\,\text{N}}$

**EVALUATE**

Positive: to the right, the way the wall pushes. The change is $20$, not $20 - 30$: the sign reversal matters. $2000\,\text{N}$ is about $510$ times the weight ($3.92\,\text{N}$), so ignoring gravity was justified. ✓

**Question**

A $70\,\mathrm{kg}$ driver moving at $15\,\mathrm{m/s}$ is stopped in a collision. Find the average force on the driver if the stopping time is (a) $0.10\,\mathrm{s}$ against a rigid dashboard, (b) $0.30\,\mathrm{s}$ with an airbag.

**Predict first**

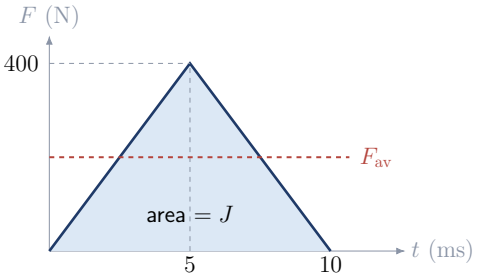
The impulse is the same in both cases. What changes the force?

**Equation (5), then (4):** $J = \Delta p = 70\,(0 - 15) = -1050\,\mathrm{N\,s}$ in both cases; $F_{\mathrm{av}} = J/\Delta t$.
(a) $F_{\mathrm{av}} = 1050/0.10 = 1.05 \times 10^4\,\mathrm{N}$. (b) $F_{\mathrm{av}} = 1050/0.30 = 3.5 \times 10^3\,\mathrm{N}$.
The same impulse delivered over three times the time gives one third of the force: that is why airbags, crumple zones and bending your knees on landing work.

*eg. a car's airbag in a crash.*

**But what if...** the force is not constant during a real impact, but rises and falls as the ball deforms? Then $\sum \vec{F}\,\Delta t$ is not enough: the impulse is the **area under the $F$–$t$ graph**.

# Impulse from a force–time graph




**Key idea**

The area under the curve is $J$. $F_{\text{av}}$ is the flat line with the same area.

**State.** For a force that varies with time, $J = \int F\,\mathrm{d}t$ is the area under the $F$–$t$ graph, from (4). The average force is $F_{\text{av}} = J/\Delta t$.

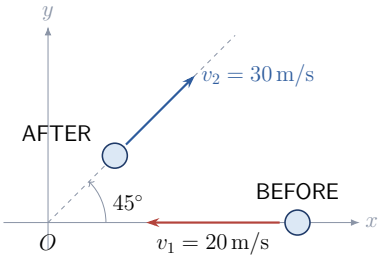
**Define.** The graph shows the force a racket exerts on a $0.057\,\text{kg}$ tennis ball, initially at rest: it rises linearly to $400\,\text{N}$ at $5\,\text{ms}$ and falls to zero at $10\,\text{ms}$. $F$ in N, $t$ in ms (convert to s before multiplying).

**Explain.**
Area (triangle): $J = \frac{1}{2}(0.010\,\text{s})(400\,\text{N}) = 2.0\,\text{N s}$.
Speed: $v = J/m = 2.0/0.057 = 35\,\text{m/s}$.
Average: $F_{\text{av}} = J/\Delta t = 2.0/0.010 = 200\,\text{N}$, half the peak.

A large force for a short time and a smaller force for a longer time deliver the same impulse if the areas match.

# Example 3: a kicked ball in two dimensions

CORE




## Key idea

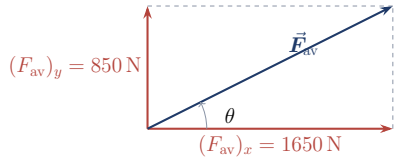
$\vec{\boldsymbol{J}} = \vec{\boldsymbol{p}}_2 - \vec{\boldsymbol{p}}_1$ is a vector difference: do each component.

## Example 3

A footballer at the Legon pitch kicks a $0.40\,\mathrm{kg}$ ball that is moving to the left at $20\,\mathrm{m/s}$. The ball leaves at $30\,\mathrm{m/s}$, $45°$ above the horizontal and to the right. Find the impulse of the net force on the ball and the average net force, if the contact time is $\Delta t = 0.010\,\mathrm{s}$.

## Predict first

Which way does $\vec{\boldsymbol{J}}$ point: mainly right, mainly up, or between? Is it $\Delta p = 30 - 20 = 10$ times $m$?




**Key idea**

$\vec{F}_{\text{av}}$ points along $\vec{J}$: its components come from $J_x$ and $J_y$.

**IDENTIFY**

Given: $m = 0.40\,\text{kg}$; $v_1 = 20\,\text{m/s}$ along $-x$; $v_2 = 30\,\text{m/s}$ at $45°$ above $+x$; $\Delta t = 0.010\,\text{s}$. Find: $\vec{J}$ and $\vec{F}_{\text{av}}$.

**SET UP**

$+x$ right, $+y$ up: $v_{1x} = -20$, $v_{1y} = 0$; $v_{2x} = 30\cos 45° = 21.2$, $v_{2y} = 30\sin 45° = 21.2\,\text{m/s}$. Use (5) per axis.

**EXECUTE**

Equations: $J_x = m(v_{2x} - v_{1x})$, $J_y = m(v_{2y} - v_{1y})$, $(F_{\text{av}})_x = J_x/\Delta t$, $(F_{\text{av}})_y = J_y/\Delta t$

Substitute: $J_x = (0.40)[21.2 - (-20)] = 16.5\,\text{N s}$; $J_y = (0.40)(21.2 - 0) = 8.5\,\text{N s}$

Components: $(F_{\text{av}})_x = 16.5/0.010 = 1650\,\text{N}$; $(F_{\text{av}})_y = 8.5/0.010 = 850\,\text{N}$

Magnitude: $F_{\text{av}} = \sqrt{(1650)^2 + (850)^2}\,\text{N}$ $\boxed{F_{\text{av}} = 1.9 \times 10^3\,\text{N}}$

Direction: $\theta = \tan^{-1}\dfrac{850}{1650}$ $\boxed{\theta = 27°\text{ above } +x}$

**EVALUATE**

Both components positive, mainly right: as predicted. $\vec{J}$ is the *vector* difference $\vec{p}_2 - \vec{p}_1$; the difference of magnitudes would give only $4.0\,\text{N s}$. ✓

# Your turn: a tennis ball and a racket

**Independent problem**

A $0.058\,\mathrm{kg}$ tennis ball arrives horizontally at $30\,\mathrm{m/s}$ and leaves the racket in the opposite direction at $45\,\mathrm{m/s}$. The contact lasts $5.0\,\mathrm{ms}$. Find the impulse and the average force on the ball.

**Predict first**

Sketch with $+x$ along the outgoing velocity. Sign of $J$? Roughly how many times the ball's weight?

**IDENTIFY, SET UP**

Given: $m = 0.058\,\mathrm{kg}$, $v_1 = -30$, $v_2 = +45\,\mathrm{m/s}$, $\Delta t = 5.0 \times 10^{-3}\,\mathrm{s}$; $+x$ along the outgoing velocity. Use (5), (4).

**EXECUTE**

$J = m(v_2 - v_1) = (0.058)[45 - (-30)]$ $\boxed{J = 4.4\,\mathrm{N\,s}}$ along the outgoing direction

$F_{\mathrm{av}} = J/\Delta t = 4.35/0.0050$ $\boxed{F_{\mathrm{av}} = 8.7 \times 10^2\,\mathrm{N}}$

**EVALUATE**

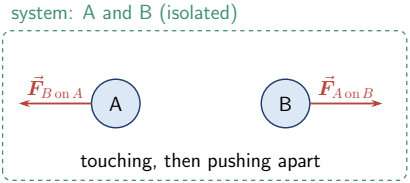
Positive in the chosen direction. About $1500$ times the ball's weight ($0.57\,\mathrm{N}$). ✓

## Common mistakes and the correct habit

| Mistake | Instead |
|---|---|
| Treating momentum as a scalar, or ignoring signs. | $\vec{\boldsymbol{p}} = m\vec{\boldsymbol{v}}$ is a vector: choose axes and keep signs. |
| Adding magnitudes of momenta pointing in different directions. | Add components: $P_x = \sum p_x$, $P_y = \sum p_y$. |
| Confusing momentum with kinetic energy. | $p$ can be negative; $K = p^2/2m$ is a scalar $\geq 0$. |
| Taking the impulse of a bounce as $mv$. | Reversing the velocity gives $\Delta p = 2mv$. |
| Calling the impulse a force, or the average force the peak force. | $J = \Delta p$ (N s); $F_{\text{av}} = J/\Delta t$. |
| Using the impulse of one force instead of the net force. | $\vec{\boldsymbol{J}}$ is the impulse of the *net* external force. |

**But what if...** the force on one object is exerted by a second object, which feels an equal and opposite force for the same time? Then the two impulses cancel in the total: conservation of momentum, next.

# A new question: from one object to two




## Key idea

Each pushes the other. Nothing outside pushes either of them.

## Predict first

Which law connects the push on $A$ to the push on $B$? What does it say about $\Delta\vec{p}_A$ and $\Delta\vec{p}_B$?

## Earlier today

One object: the net external force over a time gives $\vec{J} = \Delta\vec{p}$.

## Poll

Two astronauts, $A$ ($60\,\mathrm{kg}$) and $B$ ($90\,\mathrm{kg}$), float at rest in space and touch. They push each other apart. What is the total momentum afterwards? Which astronaut moves faster?

**But what if...** we must first say which objects form the system, and which forces are inside it and which act from outside? The next frame fixes the vocabulary.

# Vocabulary: system, internal and external forces

CORE

| Term | What it is | What it tells you | How to picture it |
|---|---|---|---|
| **System** | the objects whose momentum we add up | where to draw the boundary | a dashed box round them |
| **Internal forces** | the forces that the particles of the system exert on each other | they come in third-law pairs, so they cancel in the total | arrows between the objects |
| **External forces** | the forces exerted on any part of the system by some object outside it | only these can change the total momentum | arrows crossing the boundary |
| **Isolated system** | a system on which no external forces act | its total momentum is constant | a dashed box with no arrows across |

**Apply.** Each astronaut pushes the other (internal forces); nothing outside pushes (no external force): an *isolated* system. The total momentum is $\vec{\boldsymbol{P}} = \vec{\boldsymbol{p}}_A + \vec{\boldsymbol{p}}_B$, (3).

# Why momentum is conserved

**State.** For two bodies that interact only with each other, the forces form a third-law pair,

$$\vec{F}_{\text{A on B}} = -\vec{F}_{\text{B on A}} \quad \text{so} \quad \vec{F}_{\text{B on A}} + \vec{F}_{\text{A on B}} = 0.$$

**Define.** $\vec{F}_{\text{A on B}}$ is the force that $A$ exerts on $B$ (N), $\vec{F}_{\text{B on A}}$ the force $B$ exerts on $A$. $\vec{p}_A$, $\vec{p}_B$ are the momenta of the two bodies. $\vec{P} = \vec{p}_A + \vec{p}_B$ is the total.

**Explain.**

**1.** Each force is the net external force on its body: $\vec{F}_{\text{A on B}} = \mathrm{d}\vec{p}_B/\mathrm{d}t$ and $\vec{F}_{\text{B on A}} = \mathrm{d}\vec{p}_A/\mathrm{d}t$, from (2).

**2.** Add them:

$$\frac{\mathrm{d}\vec{p}_A}{\mathrm{d}t} + \frac{\mathrm{d}\vec{p}_B}{\mathrm{d}t} = \frac{\mathrm{d}(\vec{p}_A + \vec{p}_B)}{\mathrm{d}t} = \frac{\mathrm{d}\vec{P}}{\mathrm{d}t} = 0.$$

**3.** So $\vec{P}$ does not change in time: $\vec{P} =$ constant. The two impulses are equal and opposite, $\Delta\vec{p}_A = -\Delta\vec{p}_B$.

*Reuse:* Lecture 3 (6), the third law

**Key idea**

The momentum of each body changes. The total does not, because the internal forces cancel in pairs.

**Any number of bodies**

The argument works for any number of particles: all the internal forces cancel in pairs, so only external forces can change $\vec{P}$. With several particles, $\sum \vec{F}_{\text{ext}} = \mathrm{d}\vec{P}/\mathrm{d}t$.

**But what if...** the argument holds for any system, not just two astronauts? State it once as a rule, with the condition on the external forces.

**Rule: conservation of momentum**

$$\vec{\boldsymbol{P}}_{\text{before}} = \vec{\boldsymbol{P}}_{\text{after}} \qquad \text{(isolated system)} \qquad (6)$$

**Define.** $\vec{\boldsymbol{P}} = \vec{\boldsymbol{p}}_A + \vec{\boldsymbol{p}}_B + \cdots$ is the total momentum of the system (kg m/s), a vector. **Isolated:** the vector sum of the external forces is zero.

**Explain.** Internal forces change the momenta of the parts, never the total. Momentum is a vector, so (6) holds **separately along each axis**: $P_x$ is constant and $P_y$ is constant. If the external forces are not zero, but are much smaller than the internal forces during a short interaction, (6) still holds just before and just after: the **impulse approximation**.

**Predict first**

A skater on frictionless ice throws a heavy ball forward. Predict which way she moves, and whether the total momentum of skater and ball changes.

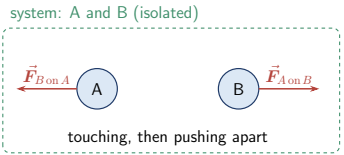
She moves backward; $\vec{\boldsymbol{P}} = 0$ before, so it is $0$ after: her momentum is equal and opposite to the ball's.

**Valid when**

the external forces on the system sum to zero, or are negligible during the interaction (a collision). A rough floor under a slowly sliding block is *not* negligible.

# Example 4: two astronauts push apart (the poll answered)

CORE




**Key idea**

$\vec{\boldsymbol{P}} = 0$ before, so $\vec{\boldsymbol{P}} = 0$ after: equal and opposite momenta.

## IDENTIFY

Given: $m_A = 60\,\text{kg}$, $m_B = 90\,\text{kg}$, both at rest before.
Find: the total momentum after; the ratio of the speeds.

## SET UP

System: $A$ and $B$. No external force, so (6) holds. $+x$ along $A$'s motion; $P_{x,\text{before}} = 0$.

## EXECUTE

Equation: $P_{x,\text{before}} = P_{x,\text{after}}$, so $0 = m_A v_{Ax} + m_B v_{Bx}$
Substitute: $0 = (60\,\text{kg})\, v_{Ax} + (90\,\text{kg})\, v_{Bx}$
Solve: $\dfrac{v_{Ax}}{v_{Bx}} = -\dfrac{90}{60}$

$|v_A| = 1.5\,|v_B|$, opposite directions

Total after: $\vec{\boldsymbol{P}}_{\text{after}} = 0$

## EVALUATE

The lighter astronaut moves $1.5$ times faster, the other way. $\Delta\vec{\boldsymbol{p}}_A = -\Delta\vec{\boldsymbol{p}}_B$ (third law). The total momentum is zero, the kinetic energy is not. ✓

**Decide first, with a reason for each, then check**

1. Can a single object have momentum but zero kinetic energy? Or kinetic energy but zero momentum?
2. Two objects move apart, yet their total momentum is zero. Is that possible?
3. Is the total momentum of two colliding cars conserved even though friction acts on the road?

**Answers**

**1. No, for a single object.** $p = 0$ means $v = 0$, so $K = 0$ too. For a *system*, the total momentum can vanish while the parts move.
**2. Yes.** Equal and opposite momenta, as with the astronauts.
**3. Yes, during the collision.** The forces between the cars are far larger than the road friction, so the impulse approximation holds.

**But what if...** we now have the law and have tested the idea? We need a routine for applying it, one that fixes the system, the axes and the signs.

### METHOD – Using $\vec{P}_{\text{before}} = \vec{P}_{\text{after}}$

**1. SYSTEM:** choose the objects so that the important forces are *internal*.
**2. ISOLATED?** Check that the external forces sum to zero, or are negligible during a brief interaction.
**3. SKETCH** before and after, with $+x$ and each velocity labelled with its sign.
**4. WRITE** one equation per axis:
$m_A v_{A1x} + m_B v_{B1x} = m_A v_{A2x} + m_B v_{B2x}$.
**5. TYPE:** stuck together $\Rightarrow$ one common $v_2$; elastic $\Rightarrow$ add $K_{\text{before}} = K_{\text{after}}$; otherwise one final velocity must be given.
**6. SOLVE**, then compute $K$ before and after to classify the collision.

**WATCH OUT:** $K_{\text{after}} > K_{\text{before}}$ is impossible for an isolated collision without an energy source: look for a sign error.
**CHECK:** is momentum conserved when you substitute your answers?

### Which tool?

**Conservation (6):** *velocities after an interaction between objects*, when the external forces are zero or negligible.

**Impulse (5):** the *force or the time* of the interaction, or when a net external force acts.

### Decide: which tool?

(a) Recoil speed of a gun. (b) Force of a wall on a ball. (c) Final speed of two trolleys that couple.

**Answers:** (a) conservation, (b) impulse, (c) conservation.

**Example 5**

A marksman holds a $3.00\,\mathrm{kg}$ rifle loosely so that it can recoil freely. He fires a $5.00\,\mathrm{g}$ bullet horizontally at $300\,\mathrm{m/s}$ relative to the ground. Find the rifle's recoil velocity and the final momentum and kinetic energy of bullet and rifle.

**Predict first**

Which moves faster, the bullet or the rifle? Which has more kinetic energy? Is kinetic energy conserved?

# Example 5: recoil of a rifle, solution

PRACTICE

**IDENTIFY**

Given: $m_r = 3.00\,\text{kg}$, $m_b = 0.00500\,\text{kg}$, $v_b = +300\,\text{m/s}$, both at rest before. Find: $v_r$; the momenta and $K_r$, $K_b$.

**SET UP**

System: rifle and bullet; the external horizontal force is negligible during the shot, so (6) holds. $+x$ along the bullet's velocity; $P_{x,\text{before}} = 0$.

**EXECUTE**

Equation: $0 = m_b v_b + m_r v_r$, so $v_r = -m_b v_b / m_r$

Substitute: $v_r = -\dfrac{(0.00500\,\text{kg})(300\,\text{m/s})}{3.00\,\text{kg}}$ $\boxed{v_r = -0.500\,\text{m/s}}$

Momenta: $p_b = +1.50\,\text{kg m/s}$, $p_r = -1.50\,\text{kg m/s}$; $K_b = \frac{1}{2}(0.00500)(300)^2 = 225\,\text{J}$, $K_r = \frac{1}{2}(3.00)(0.500)^2 = 0.375\,\text{J}$

**EVALUATE**

The momenta are equal and opposite, but the light bullet carries almost all the kinetic energy ($K = p^2/2m$). Kinetic energy rose from $0$ to about $225\,\text{J}$: it came from the chemical energy of the powder. Momentum is conserved; kinetic energy need not be. ✓

# What is a collision?

## Definition and rule: collision

A **collision** is any strong interaction between objects that lasts a relatively short time. In any collision in which the external forces can be ignored, **momentum is conserved**: (6).

**Define.** A collision includes car accidents, balls on a billiard table, neutrons hitting nuclei and a spacecraft passing Saturn. Before and after mean just before the contact starts and just after it ends.

**Explain.** If the forces between the colliding objects are much larger than any external force, we treat the objects as an isolated system. Two cars colliding on dry pavement qualify: the forces between the cars far exceed the friction of the road. **Kinetic energy** is conserved only in a special class of collisions, defined next.

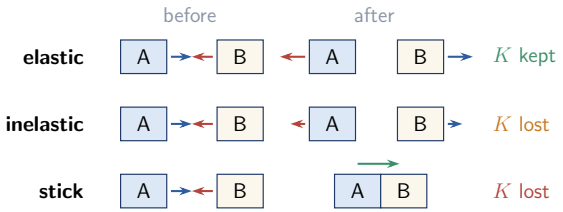
## Key idea

Momentum is the first tool for any collision: it is conserved whatever happens to the energy.

## Predict first

The system in Example 5 gained $225\,\mathrm{J}$ of kinetic energy; gliders that stick will lose some. Can kinetic energy do both?

**But what if...** the kinetic energy is sometimes kept, lost, or gained? Sorting collisions by what happens to $K$ gives their types.




## Key idea

Momentum is conserved in all three. Only the kinetic energy tells them apart.

## Common error

"Momentum is lost in an inelastic collision." It is not. Kinetic energy is partly converted into other forms: deformation, heat, sound.

**State. Elastic:** the total kinetic energy after equals that before; the forces between the objects are conservative. **Inelastic:** the total kinetic energy after is less than before. **Completely inelastic:** an inelastic collision in which the objects stick and move as one.

**Define.** $K = \frac{1}{2}mv^2$, summed over the objects (J). *Reuse:* Lecture 4 (4)

**Explain.** Billiard balls squash a little and spring back: almost perfectly elastic. Gliders with spring bumpers store $K$ as elastic potential energy while compressed, then return it. A bullet embedding in wood is inelastic. Perfectly elastic collisions between large objects are rare.

*eg. a pool table, where the balls are almost perfectly elastic.*

## Decide first, then check

For each situation, say whether the collision is elastic or inelastic, and if inelastic whether it is completely inelastic.
(a) You drop a ball from your hand. It collides with the floor and bounces back so that it just reaches your hand.
(b) You drop a different ball and it bounces back to half the height from which it was dropped.
(c) You drop a ball of clay. When it collides with the floor, it stops.

## Answers

**(a) Elastic.** All the potential energy at the start returns, so no kinetic energy is lost.
**(b) Inelastic.** The ball ends with less potential energy, so some kinetic energy was lost in the bounce.
**(c) Completely inelastic.** The ball loses all its kinetic energy and sticks to the floor.

**Common mistakes and the correct habit**

| Mistake | Instead |
|---|---|
| Conserving momentum when an external force is not negligible. | Check that the system is isolated, or that the interaction is brief. |
| Saying momentum is lost in an inelastic collision. | Momentum is kept; kinetic energy is lost. |
| Saying kinetic energy is conserved in every collision. | Only in an elastic collision. |
| Mixing the signs of the velocities. | Sketch before and after with signed velocities. |

**Take away:** the impulse of the net external force is $\Delta\vec{p}$; in an isolated system $\vec{P}$ is constant; the kinetic energy decides the type of collision.

### Contact A: what you can now do

- Calculate momentum and impulse as vectors, and find an average collision force from $\vec{\boldsymbol{J}} = \Delta\vec{\boldsymbol{p}}$.
- Say when the total momentum is conserved: the external forces sum to zero, or are negligible during the interaction.
- Use $\vec{\boldsymbol{P}}_{\text{before}} = \vec{\boldsymbol{P}}_{\text{after}}$ along each axis, and sort collisions as elastic, inelastic or completely inelastic by the kinetic energy.

### Predict first

**Before Wednesday, predict:** two gliders on a frictionless air track, $0.50\,\text{kg}$ at $+2.0\,\text{m/s}$ and $0.30\,\text{kg}$ at $-2.0\,\text{m/s}$, collide, and the lighter one leaves at $+2.0\,\text{m/s}$. Does the heavier one keep going or rebound? Is kinetic energy conserved? What changes if they stick together?

**But what if...** the collision is along a line and we must find the velocities afterwards, or the objects leave at angles? → **collisions in one dimension, then two dimensions, restitution and a swing**.

# Hope Donglo

Room 24, Department of Physics

hdonglo@ug.edu.gh


Office hours: Wednesdays, 10:00 to 12:00

Personal page: hhdonglo.github.io/phys143.html

## Contact B: Solve, then Extend

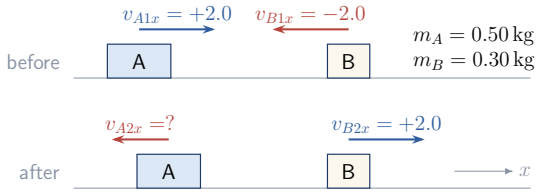
*How do we solve collisions along a line and in two dimensions, and what do they tell us about elasticity and systems?*

**Predict first**

1. A $0.40\,\mathrm{kg}$ ball hits a wall at $30\,\mathrm{m/s}$ to the left and rebounds at $20\,\mathrm{m/s}$ to the right. Find the impulse on the ball. $+x$ to the right:
   $J_x = m(v_{2x} - v_{1x}) = (0.40)[20 - (-30)] = +20\,\mathrm{N\,s}$, to the right.
2. Two astronauts, $60\,\mathrm{kg}$ and $90\,\mathrm{kg}$, push apart from rest. Which moves faster, and what is the total momentum? The lighter one, $1.5$ times faster, the other way; $\vec{\boldsymbol{P}} = 0$ before and after (isolated system).
3. In a collision of two gliders on a frictionless track, which is conserved, the total momentum or the total kinetic energy? The total momentum always; the total kinetic energy only if the collision is elastic.

# Example 6: gliders colliding along a straight line

CORE




## Key idea

Momentum is conserved along $x$. The unknown is $v_{A2x}$.

## Example 6

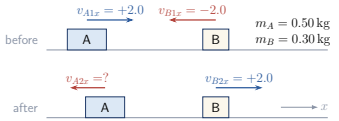
In a physics laboratory at the University of Ghana, two gliders move toward each other on a frictionless linear air track. Glider $A$ ($0.50\,\mathrm{kg}$) moves at $+2.0\,\mathrm{m/s}$ and glider $B$ ($0.30\,\mathrm{kg}$) at $-2.0\,\mathrm{m/s}$. After they collide, $B$ moves away at $+2.0\,\mathrm{m/s}$. (a) What is the final velocity of $A$? (b) How do the changes in momentum and in velocity compare? Is the collision elastic?

## Predict first

Does $A$ rebound or keep going? Which glider's velocity changes more? Is $K$ conserved?

# Example 6: gliders colliding, solution

CORE




**Key idea**

The momentum changes are equal and opposite. The velocity changes are not.

**IDENTIFY**

Given: $m_A = 0.50$, $v_{A1x} = +2.0$; $m_B = 0.30$, $v_{B1x} = -2.0$, $v_{B2x} = +2.0$ (kg, m/s). Find: $v_{A2x}$; $\Delta p$, $\Delta v$; the type.

**SET UP**

Weights and normal forces cancel; no friction: isolated along $x$. Use (6), $+x$ to the right.

**EXECUTE**

Equation: $m_A v_{A1x} + m_B v_{B1x} = m_A v_{A2x} + m_B v_{B2x}$; before: $P_x = (0.50)(2.0) + (0.30)(-2.0) = 0.40\,\text{kg m/s}$

Solve: $v_{A2x} = \dfrac{P_x - m_B v_{B2x}}{m_A} = \dfrac{0.40 - (0.30)(2.0)}{0.50}$

$$\boxed{v_{A2x} = -0.40\,\text{m/s}}$$

$\Delta p_A = 0.50(-0.40 - 2.0) = -1.2$,
$\Delta p_B = 0.30(2.0 + 2.0) = +1.2\,\text{kg m/s}$; $\Delta v_A = -2.4$,
$\Delta v_B = +4.0\,\text{m/s}$
$K_{\text{before}} = \frac{1}{2}(0.50)(2.0)^2 + \frac{1}{2}(0.30)(2.0)^2 = 1.6\,\text{J}$;
$K_{\text{after}} = \frac{1}{2}(0.50)(0.40)^2 + \frac{1}{2}(0.30)(2.0)^2 = 0.64\,\text{J}$

**EVALUATE**

$A$ rebounds, as predicted. $\Delta p_A = -\Delta p_B$, but the lighter glider's velocity changes more. $K$ fell: **inelastic**, not completely. ✓

**Rule: completely inelastic (1D)**

$$m_A v_{A1x} + m_B v_{B1x} = (m_A + m_B)\, v_{2x} \qquad (7)$$

**Define.** $v_{2x}$ is the common final velocity (m/s) of the two objects, which stick together: $v_{A2x} = v_{B2x} = v_{2x}$. Other symbols as in (6).

**Explain.** Equation (7) is (6) with one unknown, the common velocity. Solving, $v_{2x} = \dfrac{m_A v_{A1x} + m_B v_{B1x}}{m_A + m_B}$. The objects stay together, so the relative speed after is zero: the largest loss of kinetic energy possible for these initial velocities, but never all of it unless $P_x = 0$. The final velocity points the way of the larger initial momentum.

**Predict first**

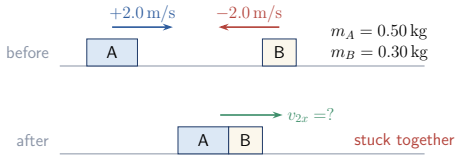
Gliders $A$ ($0.50\,\mathrm{kg}$, $+2.0\,\mathrm{m/s}$) and $B$ ($0.30\,\mathrm{kg}$, $-2.0\,\mathrm{m/s}$) stick together. Which way do they move? Faster or slower than $2.0\,\mathrm{m/s}$?

**Key idea**

Sticking together is the simplest case: one common velocity from one equation, then check $K$.

**Common error**

Assuming $K$ is conserved because the objects stick. It is not: always compute $K_{\text{before}}$ and $K_{\text{after}}$.




## Key idea

One common $v_{2x}$. Kinetic energy is always lost.

## Example 7

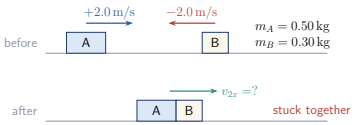
The same two gliders on the frictionless air track, $A$ $(0.50\,\mathrm{kg}, +2.0\,\mathrm{m/s})$ and $B$ $(0.30\,\mathrm{kg}, -2.0\,\mathrm{m/s})$, now have Velcro on their bumpers, so they stick together. (a) Find the final velocity. (b) Compare the initial and final kinetic energies.

## Predict first

Is the final speed more or less than $2.0\,\mathrm{m/s}$? Is most of the kinetic energy kept or lost?

# Example 7: gliders that stick together, solution

CORE




**Key idea**

One common $v_{2x}$. Kinetic energy is always lost.

**IDENTIFY**

Given: $m_A = 0.50\,\text{kg}$, $v_{A1x} = +2.0$; $m_B = 0.30\,\text{kg}$, $v_{B1x} = -2.0\,\text{m/s}$; they stick. Find: $v_{2x}$; $K_{\text{before}}$ and $K_{\text{after}}$.

**SET UP**

Isolated along $x$; $+x$ right. Use (7).

**EXECUTE**

Equation: $v_{2x} = \dfrac{m_A v_{A1x} + m_B v_{B1x}}{m_A + m_B}$

Substitute: $v_{2x} = \dfrac{(0.50)(2.0) + (0.30)(-2.0)}{0.50 + 0.30}$

Solve: $v_{2x} = \dfrac{1.0 - 0.60}{0.80}$ $\boxed{v_{2x} = +0.50\,\text{m/s}}$

Before: $K_A = \frac{1}{2}(0.50)(2.0)^2 = 1.0\,\text{J}$, $K_B = \frac{1}{2}(0.30)(2.0)^2 = 0.60\,\text{J}$; $K_{\text{before}} = 1.6\,\text{J}$

After: $K_{\text{after}} = \frac{1}{2}(0.80)(0.50)^2$ $\boxed{K_{\text{after}} = 0.10\,\text{J}}$

**EVALUATE**

To the right, slower than $2.0\,\text{m/s}$, as predicted. $K$ fell from $1.6$ to $0.10\,\text{J}$: $1.5\,\text{J}$ lost, $94\%$. Momentum is conserved ($0.40 = 0.80 \times 0.50$); the kinetic energy went into deformation and heat. ✓

**Independent problem**

A $0.40\,\text{kg}$ glider moving at $+3.0\,\text{m/s}$ collides head on with a $0.60\,\text{kg}$ glider moving at $-1.0\,\text{m/s}$. After the collision the $0.60\,\text{kg}$ glider moves at $+1.5\,\text{m/s}$. Find the velocity of the $0.40\,\text{kg}$ glider, and classify the collision.

**Predict first**

Will the $0.40\,\text{kg}$ glider rebound, or continue forward? Is $K$ conserved?

**IDENTIFY, SET UP**

Given: $m_1 = 0.40$, $v_1 = +3.0$; $m_2 = 0.60$, $v_2 = -1.0$, $v_2' = +1.5$. Find $v_1'$ and the type. Isolated along $x$: (6).

**EXECUTE**

Equation: $m_1 v_1 + m_2 v_2 = m_1 v_1' + m_2 v_2'$; substitute: $(0.40)(3.0) + (0.60)(-1.0) = 0.40\,v_1' + (0.60)(1.5)$

$0.60 = 0.40 v_1' + 0.90$ $\boxed{v_1' = -0.75\,\text{m/s}}$ (it rebounds)

$K_{\text{before}} = \frac{1}{2}(0.40)(3.0)^2 + \frac{1}{2}(0.60)(1.0)^2 = 2.1\,\text{J}$; $K_{\text{after}} = \frac{1}{2}(0.40)(0.75)^2 + \frac{1}{2}(0.60)(1.5)^2 = 0.79\,\text{J}$

**EVALUATE**

$K$ decreased and momentum is conserved: **inelastic** (not completely). ✓

# Elastic collisions in one dimension: the derivation

CORE

**State.** Object $A$ hits object $B$, at rest, head on, and no kinetic energy is lost. Two conservation laws give two equations:

$$\tfrac{1}{2}m_A v_{A1x}^2 = \tfrac{1}{2}m_A v_{A2x}^2 + \tfrac{1}{2}m_B v_{B2x}^2 \qquad (8)$$

$$m_A v_{A1x} = m_A v_{A2x} + m_B v_{B2x} \qquad (9)$$

**Define.** $v_{A1x}$ is the initial velocity of $A$; $v_{A2x}$, $v_{B2x}$ the final velocities (m/s, signed). $B$ starts at rest.

## Valid when

the collision is **elastic** and **head on**, and $B$ starts **at rest**. Dividing needs $v_{B2x} \neq 0$ and $v_{A1x} \neq v_{A2x}$: true whenever the objects collide.

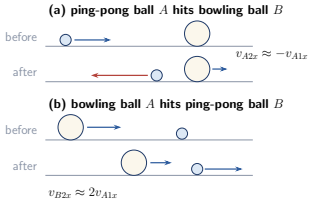
**Explain.** We want $v_{A2x}$ and $v_{B2x}$ from the masses and $v_{A1x}$.

1. **Rearrange** (8) and (9):
$$m_B v_{B2x}^2 = m_A(v_{A1x} - v_{A2x})(v_{A1x} + v_{A2x})$$
$$m_B v_{B2x} = m_A(v_{A1x} - v_{A2x})$$
2. **Divide** the first by the second:
$$v_{B2x} = v_{A1x} + v_{A2x}$$
3. **Substitute** into the second:
$m_B(v_{A1x} + v_{A2x}) = m_A(v_{A1x} - v_{A2x})$, so
$$v_{A2x} = \frac{m_A - m_B}{m_A + m_B}\, v_{A1x}, \qquad v_{B2x} = \frac{2m_A}{m_A + m_B}\, v_{A1x}$$

## Key idea

Two unknowns, two equations: momentum and kinetic energy. Dividing removes the squares.

# Elastic collision, target at rest: the result




## Key idea

The velocities depend only on the mass ratio. Light on heavy bounces back; heavy on light pushes ahead.

## Rule: elastic, target at rest

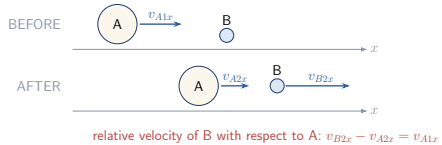
$$v_{A2x} = \frac{m_A - m_B}{m_A + m_B}\, v_{A1x} \tag{10a}$$

$$v_{B2x} = \frac{2m_A}{m_A + m_B}\, v_{A1x} \tag{10b}$$

**Define.** Symbols as in (8) and (9). Valid for an elastic head-on collision, $B$ initially at rest.

**Explain.** Special cases, each of which you can check:
**Equal masses:** $v_{A2x} = 0$, $v_{B2x} = v_{A1x}$. $A$ stops dead and gives all its momentum and kinetic energy to $B$: familiar to every pool player.
$m_A \ll m_B$ **(ping-pong ball on a bowling ball):** $v_{A2x} \approx -v_{A1x}$, $v_{B2x} \approx 0$.
$m_A \gg m_B$ **(bowling ball on a ping-pong ball):** $v_{A2x} \approx v_{A1x}$, $v_{B2x} \approx 2v_{A1x}$.
$v_{B2x} > 0$ always; $v_{A2x}$ has the sign of $m_A - m_B$.

# Elastic collisions and relative velocity




## Key idea

After the collision $B$ moves away from $A$ at the speed with which $A$ approached.

## Predict first

(10a), (10b) with $m_A = 3.0$, $m_B = 1.0\,\text{kg}$, $v_{A1x} = 4.0\,\text{m/s}$ give $v_{A2x} = 2.0$, $v_{B2x} = 6.0\,\text{m/s}$. Predict the relative velocity before and after. *Answer:* $-4.0$ before, $+4.0\,\text{m/s}$ after: it reverses.

## Rule: relative velocity, elastic

$$v_{B2x} - v_{A2x} = -\left(v_{B1x} - v_{A1x}\right) \qquad (11)$$

**Define.** $v_B - v_A$ is the velocity of $B$ relative to $A$ (m/s, signed); subscripts 1 and 2: before and after.

**Explain.** Step 2 of the derivation gave $v_{B2x} - v_{A2x} = v_{A1x}$. With $B$ at rest before, the relative velocity was $v_{B1x} - v_{A1x} = -v_{A1x}$: it **reverses**, with the same magnitude, because $A$ and $B$ approach before and separate after. Equation (11) holds for *any* straight-line elastic collision, even if both objects move at the start, and in every frame of reference.

Check: before $0 - 4.0 = -4.0$; after $6.0 - 2.0 = +4.0$. ✓

**Check the formulas**

A $m_A = 1.0\,\mathrm{kg}$ object moving at $4.0\,\mathrm{m/s}$ hits a $m_B = 3.0\,\mathrm{kg}$ target at rest, elastically and head on. Find both velocities afterwards, and verify that momentum and kinetic energy are conserved.

**EXECUTE**

(10a): $v_{A2x} = \dfrac{1.0 - 3.0}{1.0 + 3.0}(4.0) = -2.0\,\mathrm{m/s}$; (10b): $v_{B2x} = \dfrac{2(1.0)}{1.0 + 3.0}(4.0) = +2.0\,\mathrm{m/s}$

Momentum: before $(1.0)(4.0) = 4.0$; after $(1.0)(-2.0) + (3.0)(2.0) = 4.0\,\mathrm{kg\,m/s}$. ✓

$K$: before $\frac{1}{2}(1.0)(4.0)^2 = 8.0\,\mathrm{J}$; after $\frac{1}{2}(1.0)(2.0)^2 + \frac{1}{2}(3.0)(2.0)^2 = 8.0\,\mathrm{J}$. ✓

Relative velocity: before $0 - 4.0 = -4.0$; after $2.0 - (-2.0) = +4.0\,\mathrm{m/s}$: it reversed (11). ✓

**EVALUATE**

The light object rebounds from the heavy target, as expected. All three checks pass.

# A new question: beyond a straight line



## So far today

$\vec{P}$ is conserved in every isolated collision, and $K$ decides the type. But every collision so far was along one line, and either elastic or stuck together.

## The next question

What if the objects move off at angles, or only partly recover their relative speed? What if a collision is followed by more motion, or the system has several parts?

**Physical question → tool**

objects leave at angles → momentum along each axis

how elastic is the collision? → coefficient of restitution $e$

a collision, then a swing or slide → momentum, then energy

a whole system of parts → centre of mass

**But what if...** the puck leaves at an angle? Momentum is a vector, so it is conserved *separately along each axis*: collisions in two dimensions.

## Rule: components are independent

$$m_A v_{A1x} + m_B v_{B1x} = m_A v_{A2x} + m_B v_{B2x} \quad \text{(12a)}$$

$$m_A v_{A1y} + m_B v_{B1y} = m_A v_{A2y} + m_B v_{B2y} \quad \text{(12b)}$$

**Define.** Each velocity is resolved into components, $v_x = v\cos\theta$, $v_y = v\sin\theta$, with $\theta$ from $+x$ (m/s, signed). Other symbols as in (6).

**Explain.** Momentum is a vector, so (6) is one equation for each axis: the same law as in one dimension, written twice. Each final velocity has two unknown components, so two bodies have four unknowns while (12a), (12b) give two equations. We need two more facts: for example one final speed and angle, or, if elastic, $K_{\text{before}} = K_{\text{after}}$ and one angle.
*Reuse:* Lecture 1, components

## Predict first

Puck $A$ moving along $+x$ hits an identical puck $B$ at rest and moves off at $30^\circ$ above the $x$-axis. Does $B$ move above or below the axis?

$P_y = 0$ before, so $P_y = 0$ after: $B$ moves **below**, with $m_A v_{A2y} = -m_B v_{B2y}$.

## Key idea

Two dimensions needs no new law: one equation for each axis, and care with the signs of the components.

## METHOD – Using (12a) and (12b)

**1. SKETCH** before and after, with $+x$, $+y$ and every angle measured from $+x$.
**2. RESOLVE** each velocity into components; assign signs by the quadrant. *Reuse:* Lecture 1
**3. WRITE** two equations: $P_x$ conserved and $P_y$ conserved.
**4. SOLVE** for the unknown components; then reconstruct the speed and the direction: $v = \sqrt{v_x^2 + v_y^2}$, $\theta = \tan^{-1}(v_y/v_x)$, checking the quadrant.
**5. CLASSIFY:** compute $K$ before and after.

**WATCH OUT:** the $y$ momentum before need not be zero: include every component, with its sign.
**CHECK:** do both momentum components balance when you substitute? Does the direction match the sketch?

### One-dimensional or two?

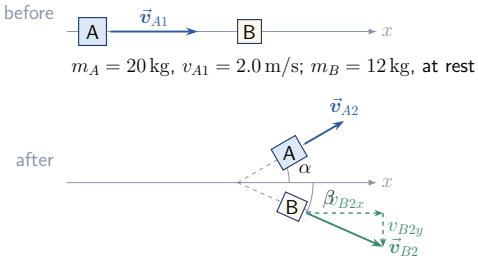
If all the velocities lie on one line, one equation is enough. If any object leaves the line, you need an equation for each axis.

### Decide: how many equations?

(a) Two gliders on a track. (b) Two pucks on an air-hockey table, one leaving at $30°$. (c) A ball dropped on a floor that does not move.

**Answers:** (a) one, (b) two, (c) one, along the vertical.

# Example 8: robots in a collision in a horizontal plane




## Key idea

$A$ leaves upward, so $B$ must leave downward. Resolve both velocities and equate the components.

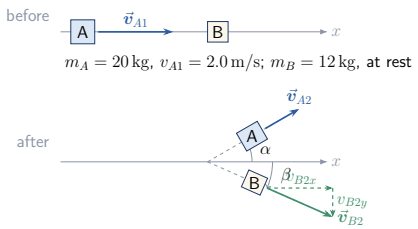
## Example 8

Two robots in a contest slide on a frictionless surface. Robot $A$ ($20\,\mathrm{kg}$) moves at $2.0\,\mathrm{m/s}$ parallel to the $x$-axis and collides with robot $B$ ($12\,\mathrm{kg}$), initially at rest. After the collision $A$ moves at $1.0\,\mathrm{m/s}$, $\alpha = 30°$ from its initial direction. What is the final velocity of $B$?

## Predict first

Is $v_{By}$ positive or negative? Is $B$ faster than $A$ at the end? Is $K$ conserved?

# Example 8: robots, solution (1): the components




**Key idea**

The $y$ momenta cancel: $(20)(0.50) = 12\,|v_{B2y}|$.

## IDENTIFY

Given: $m_A = 20\,\text{kg}$, $v_{A1} = 2.0\,\text{m/s}$ along $+x$; $m_B = 12\,\text{kg}$, at rest; $v_{A2} = 1.0\,\text{m/s}$ at $\alpha = 30^\circ$ above $+x$. Find: $\vec{v}_{B2}$.

## SET UP

Frictionless horizontal plane: isolated, so (12a), (12b) hold. $v_{A2x} = 1.0\cos 30^\circ = 0.866$, $v_{A2y} = 1.0\sin 30^\circ = 0.500\,\text{m/s}$.

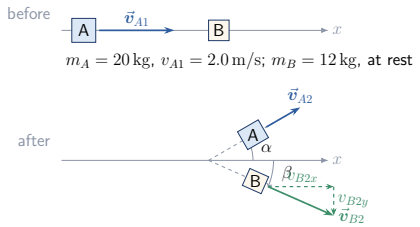
## EXECUTE

$x$: $m_A v_{A1x} = m_A v_{A2x} + m_B v_{B2x}$, so

$$v_{B2x} = \frac{(20)(2.0) - (20)(0.866)}{12} \qquad \boxed{v_{B2x} = 1.89\,\text{m/s}}$$

$y$: $0 = m_A v_{A2y} + m_B v_{B2y}$, so $v_{B2y} = -\dfrac{(20)(0.500)}{12}$

$$\boxed{v_{B2y} = -0.83\,\text{m/s}}$$




**Key idea**

$v_{B2y} < 0$: $B$ leaves below the axis, as the sketch showed.

**EXECUTE (continued)**

Magnitude: $v_{B2} = \sqrt{(1.89)^2 + (0.83)^2}\,\text{m/s}$

$\boxed{v_{B2} = 2.07\,\text{m/s}}$

Direction: $\beta = \tan^{-1}\dfrac{0.83}{1.89}$ $\boxed{\beta = 23.8^\circ \text{ below } +x}$

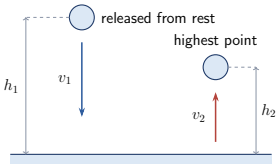
(quadrant IV)

Classify: $K_{\text{before}} = \frac{1}{2}(20)(2.0)^2 = 40\,\text{J}$;

$K_{\text{after}} = \frac{1}{2}(20)(1.0)^2 + \frac{1}{2}(12)(2.07)^2 = 35.6\,\text{J}$

**EVALUATE**

$v_{By} < 0$, as predicted. $B$ is faster than $A$. $K$ fell by $4.4\,\text{J}$: **inelastic**. Momentum check, $y$: $(20)(0.50) = 10$ and $(12)(0.83) = 10$. ✓




## Key idea

A ball dropped on a fixed floor: $e = v_2/v_1 = \sqrt{h_2/h_1}$.

## Predict first

A ball dropped from $1.0\,\mathrm{m}$ has $e = 0.50$. Predict its first rebound height. *Answer:* $h_2 = e^2 h_1 = 0.25\,\mathrm{m}$.

## Rule: coefficient of restitution

$$e = \frac{v_{B2x} - v_{A2x}}{v_{A1x} - v_{B1x}} \tag{13}$$

**Define.** $e$ is a pure number: the relative speed of separation divided by the relative speed of approach, with signed velocities along one axis.

**Explain.** Perfectly elastic and completely inelastic collisions are few in practice; $e$ measures where a real collision lies. $e = 1$: elastic, (11). $e = 0$: completely inelastic, the objects stay together ($v_{B2x} = v_{A2x}$). $0 < e < 1$: inelastic. For a ball on a fixed floor $v = \sqrt{2gh}$, so $e = \sqrt{h_2/h_1}$. *Reuse:* Lecture 5 (4)

**Example 9**

(a) A ball is dropped from a height $h_1 = 1.50\,\mathrm{m}$ onto a hard floor and rebounds to $h_2 = 0.96\,\mathrm{m}$. Find $e$.
(b) A $2.0\,\mathrm{kg}$ glider moving at $3.0\,\mathrm{m/s}$ hits a $1.0\,\mathrm{kg}$ glider at rest on a frictionless air track, with $e = 0.50$. Find both velocities after the collision, and classify it.

**Predict first**

(a) Is $e$ more or less than $0.50$? (b) Does the $2.0\,\mathrm{kg}$ glider stop, rebound or continue? Is $K$ conserved?

**IDENTIFY, SET UP**

(a) Given: $h_1 = 1.50\,\text{m}$, $h_2 = 0.96\,\text{m}$. Find $e$: use (13) with the floor fixed, speeds from $v = \sqrt{2gh}$.

(b) Given: $m_A = 2.0$, $v_{A1} = 3.0$; $m_B = 1.0$, $v_{B1} = 0$; $e = 0.50$. Find $v_{A2}$, $v_{B2}$. Two unknowns: (6) and (13).

**EXECUTE**

(a) $v_1 = \sqrt{2(9.80)(1.50)} = 5.4\,\text{m/s}$, $v_2 = \sqrt{2(9.80)(0.96)} = 4.3\,\text{m/s}$ $\boxed{e = v_2/v_1 = 0.80}$ (also $\sqrt{0.96/1.50}$)

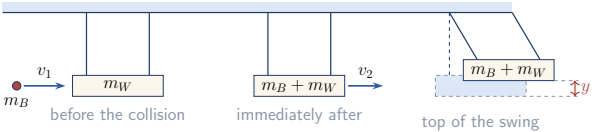
(b) (6): $(2.0)(3.0) = 2.0\,v_{A2} + 1.0\,v_{B2}$, i.e. $6.0 = 2.0v_{A2} + v_{B2}$. (13): $v_{B2} - v_{A2} = e(v_{A1} - v_{B1}) = 0.50(3.0) = 1.5$

Substitute $v_{B2} = v_{A2} + 1.5$: $6.0 = 3.0v_{A2} + 1.5$ $\boxed{v_{A2} = 1.5\,\text{m/s},\ v_{B2} = 3.0\,\text{m/s}}$

$K_{\text{before}} = \frac{1}{2}(2.0)(3.0)^2 = 9.0\,\text{J}$; $K_{\text{after}} = \frac{1}{2}(2.0)(1.5)^2 + \frac{1}{2}(1.0)(3.0)^2 = 6.75\,\text{J}$

**EVALUATE**

(a) $0.80 > 0.50$: a lively ball. (b) $A$ continues forward at half its speed; $K$ fell, $0 < e < 1$: **inelastic**. Momentum: $2.0(1.5) + 1.0(3.0) = 6.0$. ✓




## Key idea

Energy is not conserved through the impact, but it is conserved in the swing.

## Predict first

Why can we not use energy conservation from the bullet's arrival to the top of the swing? What *is* conserved in the impact? *Answer:* the impact loses kinetic energy to deformation and heat; momentum is conserved.

## Rule: two stages

**Stage 1** (the collision): momentum conserved; kinetic energy **not**. **Stage 2** (the swing): mechanical energy conserved.

**Explain.** The bullet embeds in the block: a completely inelastic collision, so $K$ is lost to deformation and heat. Just after the impact they move together at $v_2$. In the swing the tension does no work, so $K + U$ is conserved: energy conservation applies to the swing only. *Reuse:* Lecture 5 (4)

*eg. an old laboratory method for the speed of a bullet.*

**METHOD – Two stages, linked by the speed just after the collision**

**1. SPLIT** the motion at the instant the collision ends. Name the speed just after, $v_2$.
**2. STAGE 1 (collision):** momentum conservation (6), with the objects stuck or the velocities given. Do *not* use $K$ conservation unless the collision is elastic.
**3. STAGE 2 (motion after):** energy: $K_1 + U_1 + W_{\text{other}} = K_2 + U_2$ with friction or a spring if present. *Reuse:* Lecture 5 (7)
**4. LINK** through $v_2$, working backwards from what is asked.

**WATCH OUT:** using one equation (energy or momentum) across both stages.
**CHECK:** is the kinetic energy lost in stage 1 consistent with an inelastic collision?

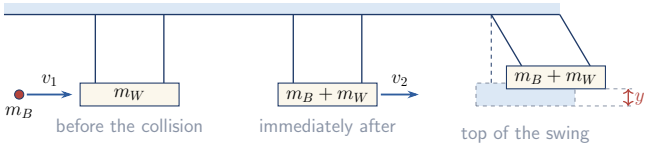
**Which stage needs which law?**

Collision: momentum. Swing, slide or spring compression: energy, with $W_{\text{other}}$ for friction.

**Decide: which law, which stage?**

A bullet embeds in a block on a rough floor. (a) Block's speed just after the impact. (b) How far it slides.

**Answers:** (a) momentum, (b) energy with friction.




## Example 10

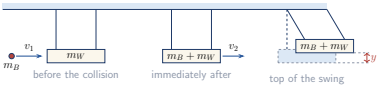
A ballistic pendulum measures the speed of a bullet. The bullet, of mass $m_B$, is fired into a block of wood of mass $m_W$ suspended like a pendulum, and makes a completely inelastic collision with it. After the impact the block swings up to a maximum height $y$. (a) Find the speed $v_1$ of the bullet in terms of $y$, $m_B$ and $m_W$. (b) Find $v_1$ and the fraction of the bullet's kinetic energy lost, for $m_B = 10\,\mathrm{g}$, $m_W = 2.0\,\mathrm{kg}$, $y = 0.20\,\mathrm{m}$.

## Predict first

Is the bullet speed of order $10^1$, $10^2$ or $10^3\,\mathrm{m/s}$? Is most of the kinetic energy lost?

# Example 10: measuring a bullet's speed, solution

CORE




**IDENTIFY, SET UP**

Given: symbols $m_B$, $m_W$, $y$; then $m_B = 0.010\,\text{kg}$, $m_W = 2.0\,\text{kg}$, $y = 0.20\,\text{m}$. Find $v_1$; the fraction of $K$ lost. Two stages: collision, momentum (6); swing, energy. *Reuse:* Lecture 5 (4)

**EXECUTE**

Stage 2 (swing): $\frac{1}{2}(m_B + m_W)v_2^2 = (m_B + m_W)\,g\,y$, so $v_2 = \sqrt{2gy}$

Stage 1 (impact): $m_B v_1 = (m_B + m_W)v_2$, so (14)

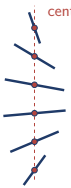
$$\boxed{v_1 = \frac{m_B + m_W}{m_B}\sqrt{2gy}}$$

(b) $v_2 = \sqrt{2(9.80)(0.20)} = 1.98\,\text{m/s}$; $\quad v_1 = \dfrac{2.01}{0.010}(1.98)$ $\quad \boxed{v_1 = 4.0 \times 10^2\,\text{m/s}}$

$K_{\text{before}} = \frac{1}{2}(0.010)(398)^2 = 790\,\text{J}$; $\quad K_{\text{after}} = \frac{1}{2}(2.01)(1.98)^2 = 3.9\,\text{J}$ $\quad \boxed{99.5\%\ \text{lost}}$

**EVALUATE**

Order $10^2\,\text{m/s}$, as predicted. The fraction lost is $m_W/(m_B + m_W) = 0.995$ whatever the speed: almost all the energy goes into deformation and heat in the collision. ✓




## Key idea

The wrench tumbles, yet its centre of mass moves in a straight line.

## Definition and rules: centre of mass

$$x_{\text{cm}} = \frac{m_1 x_1 + m_2 x_2 + \cdots}{m_1 + m_2 + \cdots} = \frac{\sum m_i x_i}{M} \tag{15}$$

$$\vec{\boldsymbol{P}} = M\vec{\boldsymbol{v}}_{\text{cm}} \qquad \sum \vec{\boldsymbol{F}}_{\text{ext}} = M\vec{\boldsymbol{a}}_{\text{cm}} \tag{16, 17}$$

**Define.** $m_i$, $x_i$: mass and position of particle $i$ (kg, m); $M = \sum m_i$; $x_{\text{cm}}$ is the position of the centre of mass (same form for $y$, $z$); $\vec{\boldsymbol{v}}_{\text{cm}}$, $\vec{\boldsymbol{a}}_{\text{cm}}$ its velocity and acceleration.

**Explain.** The mass-weighted average position: nearer the heavier object. The total momentum equals $M\vec{\boldsymbol{v}}_{\text{cm}}$, so (6) says the centre of mass of an isolated system moves at constant velocity. It moves as if all the mass were there and all the external forces acted there; internal forces cannot move it. If the system starts at rest and $\sum \vec{\boldsymbol{F}}_{\text{ext}} = 0$, it **stays where it is**. Full treatment: Lecture 8.

## Predict first

Two skaters of different mass on frictionless ice pull on a rope. Who moves farther? Does the centre of mass move? *Answer:* the lighter one moves farther; the centre of mass stays put (Example 11).



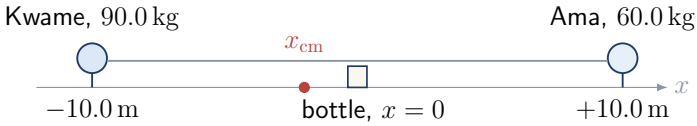

**Key idea**

Symmetry locates the centre of mass without a calculation. It need not lie inside the object.

**State.** If a homogeneous object has a geometric centre (cube, sphere, cylinder), that is where its centre of mass is. If it has an axis of symmetry (disk, donut), the centre of mass lies on that axis.

**Explain.** A donut has its centre of mass in the hole, outside the material. For an object like a wrench, with no symmetry, (15) applies in each coordinate. Example: $2.0\,\mathrm{kg}$ at $x = 0$ and $6.0\,\mathrm{kg}$ at $x = 4.0\,\mathrm{m}$ give $x_{\mathrm{cm}} = \dfrac{0 + (6.0)(4.0)}{8.0} = 3.0\,\mathrm{m}$, nearer the heavier mass.

**But what if...** we test the prediction about the two skaters on two people, with numbers: Example 11.




**Example 11**

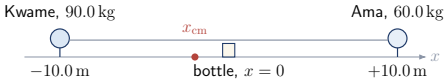
Kwame ($90.0\,\mathrm{kg}$) and Ama ($60.0\,\mathrm{kg}$) are $20.0\,\mathrm{m}$ apart on a frictionless horizontal surface. Midway between them is a bottle of water. They pull on the ends of a light rope stretched between them. When Kwame has moved $6.0\,\mathrm{m}$ toward the bottle, how far and in what direction has Ama moved?

**Predict first**

Who moves farther? Where do they meet? Does it depend on who pulls harder?

# Example 11: a tug-of-war on frictionless ground, solution

CORE




**IDENTIFY**

Given: $m_K = 90.0\,\mathrm{kg}$ at $x = -10.0\,\mathrm{m}$; $m_A = 60.0\,\mathrm{kg}$ at $x = +10.0\,\mathrm{m}$; bottle at $x = 0$; Kwame moves to $x = -4.0\,\mathrm{m}$. Find: Ama's new position $x_2$ and the distance she moved.

**SET UP**

No horizontal external force and no motion at the start: $\vec{P} = 0$, so $\vec{v}_{\mathrm{cm}} = 0$ and the centre of mass stays at rest (16). Ignore the light rope. Use (15).

**EXECUTE**

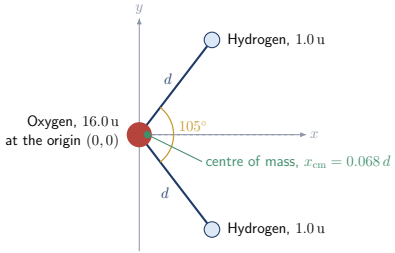
Equation: $x_{\mathrm{cm}} = \dfrac{m_K x_K + m_A x_A}{m_K + m_A}$; initially $x_{\mathrm{cm}} = \dfrac{(90.0)(-10.0) + (60.0)(10.0)}{150.0} = -2.0\,\mathrm{m}$

Substitute with the centre of mass unmoved: $\dfrac{(90.0)(-4.0) + (60.0)\,x_2}{150.0} = -2.0$

Solve: $60.0\,x_2 = -300 + 360 = 60$ $\boxed{x_2 = +1.0\,\mathrm{m}}$; she moved $10.0 - 1.0 = \boxed{9.0\,\mathrm{m}\text{ toward the bottle}}$

**EVALUATE**

The distances, $6.0$ and $9.0\,\mathrm{m}$, are in the inverse ratio of the masses, $60.0/90.0 = 2/3$. They meet at $x_{\mathrm{cm}} = -2.0\,\mathrm{m}$ and Ama reaches the bottle first, however hard either pulls: pulling harder only makes them move faster. ✓




## Example 12

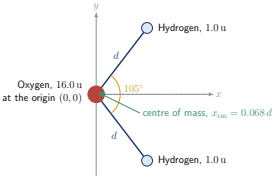
In a simple model of a water molecule the oxygen–hydrogen separation is $d = 9.57 \times 10^{-11}\,\mathrm{m}$ and the angle between the two bonds is $105^\circ$. Each hydrogen atom has mass $1.0\,\mathrm{u}$ and the oxygen atom has mass $16.0\,\mathrm{u}$. Find the position of the centre of mass.

## Predict first

Is the centre of mass nearer the oxygen or the hydrogen atoms? On which line must it lie? Is it inside the oxygen atom?

# Example 12: the centre of mass of a water molecule, solution

CORE




**Key idea**

Apply (15) to each coordinate. Symmetry fixes $y_{cm} = 0$ before any arithmetic.

**IDENTIFY**

Given: $d = 9.57 \times 10^{-11}$ m; $m_H = 1.0$ u (two atoms); $m_O = 16.0$ u; angle $105°$. Find: $x_{cm}$ and $y_{cm}$.

**SET UP**

The mass of an atom is almost all in its tiny nucleus, so treat each atom as a point. Oxygen at the origin, $x$-axis along the symmetry axis: each hydrogen is at $x = d\cos 52.5°$, $y = \pm d\sin 52.5°$. Use (15) for $x$ and $y$.

**EXECUTE**

Equation: $x_{cm} = \dfrac{m_H x_{H1} + m_H x_{H2} + m_O x_O}{m_H + m_H + m_O}$

Substitute:

$$x_{cm} = \frac{(1.0)(d\cos 52.5°) + (1.0)(d\cos 52.5°) + (16.0)(0)}{1.0 + 1.0 + 16.0} = \boxed{0.068\,d}$$

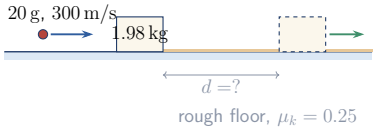
$$y_{cm} = \frac{(1.0)(d\sin 52.5°) + (1.0)(-d\sin 52.5°) + (16.0)(0)}{18.0} = \boxed{0}$$

With $d = 9.57 \times 10^{-11}$ m:

$$\boxed{x_{cm} = (0.068)(9.57 \times 10^{-11}\text{ m}) = 6.5 \times 10^{-12}\text{ m}}$$

**EVALUATE**

Much closer to the heavy oxygen than to either hydrogen, and on the symmetry axis (a $180°$ turn about that axis leaves the molecule unchanged), as predicted. The unit u cancels in the ratio. ✓



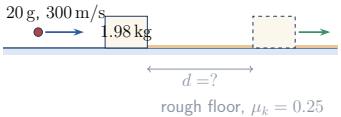

## Application problem: no method is named

A $20\,\mathrm{g}$ bullet moving horizontally at $300\,\mathrm{m/s}$ embeds in a $1.98\,\mathrm{kg}$ block at rest on a rough floor ($\mu_k = 0.25$).

(a) What is the speed of the block and bullet just after the impact?
(b) How far do they slide?
(c) What fraction of the bullet's kinetic energy is lost in the collision?

## Predict first

IDENTIFY → stages and system → PREDICT → EXECUTE → EVALUATE. Is the sliding distance more or less than $10\,\mathrm{m}$? Which law does each stage need?




**Key idea**

Two stages. Momentum in the collision, energy with friction in the slide.

**IDENTIFY**

Given: $m = 0.020\,\text{kg}$, $v = 300\,\text{m/s}$; $M = 1.98\,\text{kg}$ at rest; $\mu_k = 0.25$. Find: (a) the speed $V$ just after; (b) the distance $d$; (c) the fraction of $K$ lost.

**SET UP**

Stage 1: the impact is brief and the bullet's push is far larger than the friction on the block, so (6) holds. Stage 2: the block and bullet slide against friction: work–energy. *Reuse:* Lecture 5 (7), (8a)

**EXECUTE: (a) stage 1**

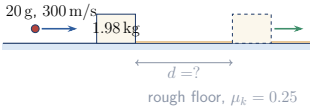
Equation: $mv = (m + M)\,V$, so $V = \dfrac{mv}{m + M}$

Substitute: $V = \dfrac{(0.020\,\text{kg})(300\,\text{m/s})}{0.020\,\text{kg} + 1.98\,\text{kg}}$ $\boxed{V = 3.0\,\text{m/s}}$

**EXECUTE: (c) the energy lost**

$K_{\text{before}} = \frac{1}{2}(0.020)(300)^2 = 900\,\text{J}$;

$K_{\text{after}} = \frac{1}{2}(2.00)(3.0)^2 = 9.0\,\text{J}$ $\boxed{99\%\ \text{lost}}$




**Key idea**

Friction removes all the $9.0\,\text{J}$: $f_k d = K$.

**EXECUTE: (b) stage 2**

Equation: $-\mu_k(m+M)g\,d = 0 - \frac{1}{2}(m+M)V^2$, so $d = \dfrac{V^2}{2\mu_k g}$

Substitute: $d = \dfrac{(3.0\,\text{m/s})^2}{2(0.25)(9.80\,\text{m/s}^2)}$ $\boxed{d = 1.8\,\text{m}}$

**EVALUATE**

$d < 10\,\text{m}$, as predicted. The heavy block moves slowly after absorbing the bullet's momentum, but almost all the energy was lost in the collision, before the slide began. Check: $f_k d = (0.25)(2.00)(9.80)(1.84) = 9.0\,\text{J} = K_{\text{after}}$. ✓

**MODIFY**

With $\mu_k = 0.50$, $d$ halves to $0.92\,\text{m}$: $V$ is set by momentum alone, the distance by friction alone.

**Which tools, and why?**

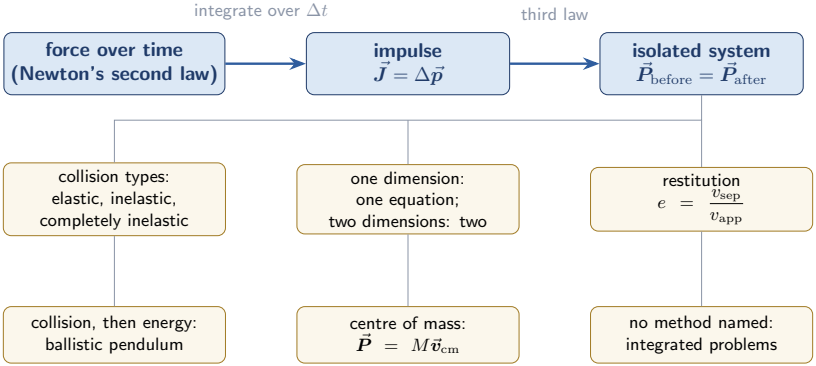
Momentum for the collision, where energy is not conserved; energy with $W_{\text{other}}$ for the slide, where momentum is not (friction is an external force).

## Common mistakes and the correct habit

| Mistake | Instead |
|---|---|
| Using one equation instead of one per axis. | Conserve $P_x$ and $P_y$ separately. |
| Using (10a), (10b) when the target is moving or the collision is not elastic. | Use (6) with (11) instead, or (6) with $K$. |
| Using energy conservation through the collision stage. | Split: momentum in the collision, energy afterwards. |
| Using a negative $e$, or $e > 1$. | $0 \leq e \leq 1$, with signed velocities in $v_{A1} - v_{B1}$. |
| Ignoring the $y$-component in a glancing collision. | Two components, two equations. |
| Saying "elastic" means no deformation. | It means no kinetic energy is lost: the deformation is temporary and recovered. |
| Letting internal forces move the centre of mass. | Only external forces change $\vec{\boldsymbol{P}}$ and $\vec{\boldsymbol{v}}_{\text{cm}}$. |

**But what if...** you can now handle a collision, a collision with angles, $e$, a collision then a swing, and a centre of mass? Pull the whole week together: the Summary.

# Week 7 concept map




**momentum: always in an isolated collision; kinetic energy: only if elastic**

A net external force acting over a time gives an impulse, which equals the change of momentum. In an isolated system the internal forces cancel, so the total momentum is constant. Kinetic energy separates the collision types.

## Key ideas

- **Momentum is a vector: $\vec{p} = m\vec{v}$. Add components. It is not kinetic energy: $K = p^2/2m$ is a scalar.**
- **The impulse of the net external force equals the change in momentum:** $\vec{J} = \int \sum \vec{F}\,\mathrm{d}t = \Delta\vec{p} = \vec{F}_{\text{av}}\Delta t$. It is not a force; it is the area under the $F$–$t$ graph.
- The same impulse over a longer time means a smaller force: airbags, crumple zones. Reversing the velocity gives $\Delta p = 2mv$.
- **In an isolated system the total momentum is conserved: $\vec{P}_{\text{before}} = \vec{P}_{\text{after}}$, one equation for each axis.** Internal forces cancel in pairs.
- **Momentum is conserved in every collision when the external forces are negligible; kinetic energy only in an elastic one.** Inelastic: $K$ decreases. Completely inelastic: the objects stick.
- **Elastic, target at rest: $v_{A2x} = \frac{m_A - m_B}{m_A + m_B} v_{A1x}$, $v_{B2x} = \frac{2m_A}{m_A + m_B} v_{A1x}$; the relative velocity reverses.**
- Two dimensions: two equations, (12a) and (12b). $e = v_{\text{sep}}/v_{\text{app}}$: 1 elastic, 0 completely inelastic.
- **Collision, then swing or slide: momentum for the collision, energy for the rest.**
- **The centre of mass: $x_{\text{cm}} = \sum m_i x_i / M$, $\vec{P} = M\vec{v}_{\text{cm}}$; internal forces cannot move it.**
- **Habits:** sketch before and after, predict, check isolated, calculate by components, check units, signs and $K_{\text{after}} \leq K_{\text{before}}$.

**Reference tools:** the next slides (equation map, master toolkit). The Reference deck holds the seven toolkits and the equation-anatomy slides.

# Reference tools: the equation map

| Idea | Equation | Conditions and units |
|---|---|---|
| **(1), (2) Momentum** | $\vec{\boldsymbol{p}} = m\vec{\boldsymbol{v}}, \quad \sum \vec{\boldsymbol{F}} = \mathrm{d}\vec{\boldsymbol{p}}/\mathrm{d}t$ | kg m/s; vector |
| **(3) System momentum** | $\vec{\boldsymbol{P}} = \vec{\boldsymbol{p}}_A + \vec{\boldsymbol{p}}_B + \cdots$ | vector sum |
| **(4), (5) Impulse** | $\vec{\boldsymbol{J}} = \int \sum \vec{\boldsymbol{F}}\,\mathrm{d}t = \vec{\boldsymbol{F}}_{\mathrm{av}}\Delta t = \Delta\vec{\boldsymbol{p}}$ | N s; net external force |
| **(6) Conservation** | $\vec{\boldsymbol{P}}_{\mathrm{before}} = \vec{\boldsymbol{P}}_{\mathrm{after}}$ | isolated; each axis |
| **(7) Completely inelastic** | $m_A v_{A1x} + m_B v_{B1x} = (m_A + m_B) v_{2x}$ | 1D; objects stick |
| **(8), (9) Elastic, target at rest** | $K$ and $P_x$ conserved | 1D, head on |
| **(10a), (10b) Elastic results** | $v_{A2x} = \frac{m_A - m_B}{m_A + m_B} v_{A1x}, \quad v_{B2x} = \frac{2m_A}{m_A + m_B} v_{A1x}$ | elastic, $B$ at rest |
| **(11) Relative velocity** | $v_{B2x} - v_{A2x} = -(v_{B1x} - v_{A1x})$ | any 1D elastic collision |
| **(12a), (12b) Two dimensions** | $P_x$ and $P_y$ conserved separately | isolated |
| **(13) Restitution** | $e = (v_{B2x} - v_{A2x})/(v_{A1x} - v_{B1x})$ | $0 \le e \le 1$ |
| **(14) Ballistic pendulum** | $v_1 = \frac{m_B + m_W}{m_B}\sqrt{2gy}$ | two stages |
| **(15)–(17) Centre of mass** | $x_{\mathrm{cm}} = \sum m_i x_i / M, \quad \vec{\boldsymbol{P}} = M\vec{\boldsymbol{v}}_{\mathrm{cm}},$ $\sum \vec{\boldsymbol{F}}_{\mathrm{ext}} = M\vec{\boldsymbol{a}}_{\mathrm{cm}}$ | m; kg m/s |

## MASTER TOOLKIT – Momentum

**1. IDENTIFY** The system, the time interval, and what is asked: a velocity, a force, a time, a distance, a position.
**2. SKETCH** Before and after (and any later stage), with axes and signed velocities.
**3. ISOLATED?** Sum of external forces zero, or negligible during a brief collision.
**4. PREDICT** Directions, speeds, and whether $K$ falls.
**5. METHOD** Impulse, conservation, collision type, $e$, or a second stage with energy.
**6. EXECUTE** One equation per axis; add $e$ or $K$ as needed.
**7. EVALUATE** Units, signs, $K_{\text{after}} \leq K_{\text{before}}$, momentum check, a second method.

## Step 5: the key move for each model

**One object, outside force:** $\vec{\boldsymbol{J}} = \Delta\vec{\boldsymbol{p}}$, $F_{\text{av}} = J/\Delta t$.
**Isolated system:** $\vec{\boldsymbol{P}}_{\text{before}} = \vec{\boldsymbol{P}}_{\text{after}}$, per axis.
**Stick together:** one common velocity (7).
**Elastic:** add $K$ conserved, or (11).
**Part elastic:** add $e$ (13).
**Collision, then motion:** momentum, then energy; link by $v_2$.
**Several parts, external forces:** centre of mass, (16), (17).

**Framework:** IDENTIFY (1), SET UP (2–5), EXECUTE (6), EVALUATE (7).

# Where this goes next

## Lecture 8: rotation

Angular momentum $\vec{\boldsymbol{L}} = \vec{\boldsymbol{r}} \times \vec{\boldsymbol{p}}$ obeys the same conservation law, and torque is the rate of change of $\vec{\boldsymbol{L}}$, as force is of $\vec{\boldsymbol{p}}$. The centre of mass returns in full.

## Lecture 10: gases

A gas exerts pressure through the impulse of molecules striking the wall; the molecular collisions are elastic, and (10a), (10b) describe them.

## Lectures 11 and 12: heat

Energy lost in an inelastic collision becomes thermal energy; the first and second laws say where it goes and why it cannot be recovered.

## Earlier weeks, now working together

Lecture 3's $\sum \vec{\boldsymbol{F}} = \mathrm{d}\vec{\boldsymbol{p}}/\mathrm{d}t$ gives impulse; the third law gives conservation. Lecture 4's $K$ and Lecture 5's energy equation handle the stage after a collision. Week 1's components make two dimensions two equations.

## Physical meaning

Momentum and energy are the two conservation laws for collisions. Momentum is always conserved when the external forces are negligible; energy tells you what is kept as motion.

- ☐ calculate momentum as a vector and distinguish it from $K$?
- ☐ find impulse from $\Delta\vec{p}$ and from a force graph?
- ☐ compute an average collision force?
- ☐ decide whether momentum is conserved?
- ☐ solve recoil and one-dimensional collisions?
- ☐ classify a collision from its kinetic energy?
- ☐ solve a two-dimensional collision by components?
- ☐ use the coefficient of restitution?
- ☐ solve a ballistic pendulum in two stages?
- ☐ find a centre of mass and state $\vec{P} = M\vec{v}_{\text{cm}}$?
- ☐ select the appropriate model in an unfamiliar problem?

**Next**

Tutorial sheet: extra impulse, conservation, collision, restitution and two-stage problems, with a ladder from independent practice to exam-style items. The Reference deck holds the material cut from the live lecture. The supplementary notes hold the impulse theorem for a variable force, the centre-of-mass frame, rocket propulsion and more on the centre of mass.

# Hope Donglo

Room 24, Department of Physics

hdonglo@ug.edu.gh


Office hours: Wednesdays, 10:00 to 12:00

Personal page: hhdonglo.github.io/phys143.html